DSP – LEC – 5 - Ch-2

CHAPTER – 2

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DISCRETE TRANSFORMS (DFT & FFT)

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2.6 Decimation-in-Time FFT Algorithm

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2.1 Discrete Fourier Transform (DFT)

2.2 Inverse Discrete Fourier Transform (IDFT)

2.3 Properties of the DFT

2.4 Computational complexity of the DFT

Ch. 2- Objectives

Calculate the frequency components, X(k), of the time components x(n). As shown in Fig. A.

1. By calculating X(k) using the DFT and calculating its inverse, x(n) using the IDFT.
2. By calculating X(k) using the FFT algorithm and calculating its inverse, x(n) using the IFFT algorithm.

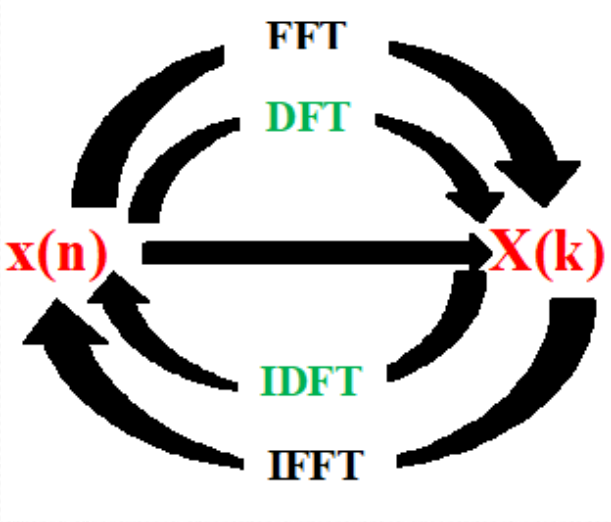


Fig. A.

Sec. 2.1

1.1 Discrete Fourier Transform (DFT)

Assume that a waveform has been sampled at regular time intervals T to produce the sample sequence

{x(nT)} = x(0), x(T), ……., x[(N-1) T]

of N sample values, where n is the sample number from n = 0 to n = N –1 .

DFT of x(nT) is defined as the sequence of complex values

{X(kΩ)} = X(0), X(Ω), …., X[(N-1) Ω]

in the frequency domain, where is the first harmonic frequency given by

Ω = 2π /NT.

Sec. 2.1 Cont. - 1

Thus X(kΩ) have real and imaginary components in general so that for the kth harmonic

$$X(k) = R(k) + jI(k) \qquad (2.1)$$

and

$$|X(K)| = [R^2(k) + I^2(k)]^{\frac{1}{2}} \qquad (2.2)$$

and X(k) has the associated phase angle

$$\Phi(k) = \tan^{-1}[I(k)/R(k)] \qquad (2.3)$$

Sec. 2.1 Cont. - 2

DFT values, X(k), are given by:

$$X(k) = F_D[x(nT)] = \sum_{n=0}^{N-1} x(nT)e^{-jk\Omega nT},\ k = 0,1,\ldots,N-1 \qquad (2.4)$$

where

F_D **.... denotes the discrete Fourier transformation and**

k..... represents the harmonic number of the transform component.

$$\Omega = 2\pi/NT. \qquad (2.4\text{-}2)$$

- **Using Eq. 2.4-2 in 2.4, we obtain:**

$$X(k) = \sum_{n=0}^{N-1} x(nT)e^{-jk2\pi n/N},\quad k=0, 1, \ldots\ldots, N-1.$$

Sec. 2.1 Cont. - 3

Find the relation between DFT and the Fourier integral* or, *Fourier transform?

Equation (2.4) can be seen to be analogous to the Fourier transform of Eq. F-1

$$F(j\omega) = \int_{-\infty}^{\infty} f(t)e^{-j\omega t}dt \qquad \text{(F-1)}$$

F(j ω) ….. is complex and is known as the *Fourier integral* or, *Fourier transform*.

when:

f(t) = 0 for t<0 and t>(N-1)T

by putting

x(nT) = f(t), kΩ = ω , and nT = t,

so the two transforms may be expected to have similar properties.

Sec. 2.1 Cont. - 4

The transforms are not, however, equal. Thus, making theses substitutions in Eq. F-1, and putting

dt = T, and replacing the integral by a summation gives

$$\sum_{n=0}^{N-1} x(nT) e^{-jk\Omega nT} T = F(j\omega) \quad \text{for } 0 \leq t \leq (N-1)T. \qquad (2.5)$$

Comparing Eq. 2.4 with 2.5 reveals that:

$$X(k) = F_D[x(nT)] = \sum_{n=0}^{N-1} x(nT) e^{-jk\Omega nT}, \; k = 0,1,\ldots, N-1 \qquad (2.4)$$

$$F(j\ ) = X(k)\ T \qquad (2.6)$$

i. e.

FT = DFT * T

showing that the FT components are related to the DFT components by the sampling interval, and may be obtained by multiplying the DFT components by the sampling interval, T.

Sec. 2.1 Cont. - 5

Example 2.1

It is appropriate at this point to illustrate the use of Eq. 2.4 by evaluating a simple case. The DFT of the sequence {1,0,0,1} will be evaluated.

In this section it is assumed that the sequence {1,0,0,1} has already been preprocessed. Assume that this data represents four consecutive voltages x(0) = 1, x(T) = 0, x(2T) =0, x(3T) = 1, recorded at time intervals, T. Thus N =4.

It is required to find the complex values X(k) for k =0, k =1, k =2 and k =3 (since N-1=3).

Sec. 2.1 Cont. -6

Solution:

Calculate X(k) using Eq. 2.4 and Ω = 2π /NT.

With k=0, Eq. 2.4 becomes

$X(0)= \sum_{n=0}^{3} x(nT)\, e^{-j0}$

$= \sum_{n=0}^{3} x(nT)$

= x(0) + x(T) + x(2T) + x(3T)

= 1+0+0+1

= 2

so X(0) = 2 is entirely real, of magnitude 2 and phase angle $\Phi(0)= 0$.

Sec. 2.1 Cont. -7

With k=1, Eq. 2.4 becomes

$X(1) = \sum_{n=0}^{3} x(nT)\, e^{-j\Omega nT}$

$X(1) = \sum_{n=0}^{3} x(nT)\, e^{-j\Omega n 2\pi/N\Omega}$

$= \sum_{n=0}^{3} x(nT)\, e^{-j2\pi n/N}$

$= 1 + 0 + 0 + 1\, e^{-j2\pi 3/4}$

$= 1 + e^{-j3\pi/2}$

$= 1 + \cos\left(\frac{3\pi}{2}\right) - j \sin\left(\frac{3\pi}{2}\right)$

$= 1 + j$

Thus

X(1) =1+j and is complex with magnitude and has angle $\Phi(\pi) = tan^{-1}(1) = 45^{o}$.

Sec. 2.1 Cont. -8

With k =2, Eq. 2.4 becomes

$X(2) = \sum_{n=0}^{3} x(nT)\, e^{-j2\Omega nT}$

$= \sum_{n=0}^{3} x(nT)\, e^{-j2n2\pi/N}$

$= \sum_{n=0}^{3} x(nT)\, e^{-j4\pi n/N}$

$= 1 + 0 + 1 + 1\, e^{-j4\pi n/N}$

$= 1 + 0 + 0 + e^{-j3\pi}$

$= 1 - 1 = 0$

Thus X(2) =0, of magnitude zero and phase angle $\Phi(2)=0$.

Finally, for k = 3, Eq. 2.4 becomes

X(3) = x(nT)e-j3n2 /N

=1 + 0 + 0 + e-j9 /2

= 1 – j

Thus X(3) = 1– j, of magnitude $\sqrt{2}$ and phase angle $\Phi(3) = -45^{o}$.

So, for the time series x(n)={1,0,0,1} has the DFT given by X(k)= {2,1 +j,0,1-j}.

Sec. 2.1 Cont. - 9

It is common practice to represent the DFT by the plots of $|X(K)|$ versus k Ω and of Φ(k) versus kΩ .

This may be done in terms of harmonics of Ω , or in terms of frequency if Ω is known.

To find it is necessary to know the value of the sampling interval, T.

If it is assumed that the above data sequence had been sampled at 8 kHz then:

$T = 1/(8 \times 10^3) = 125$ μs.

Then $\Omega = 2\pi/NT = 2\pi/(4 \times 125 \times 10^{-6}) = 1.57$ kHz.

Hence $2\Omega = 25.14$ kHz and

$3\Omega = 37.71$ kHz.

Figure 2.1(a) is a plot of x(nT) versus t,

Fig. 2.1(b) is a plot of $|X(K)|$ versus k Ω , and

Fig. 2.1(c) is a plot of Φ(k) versus k Ω .

Sec. 2.1 Cont. - 10

It is noteworthy that component, that is about harmonic number N/2, and that in Fig. 2.1(c) the phase angles are an odd function centered round this component. These results are more generally true.

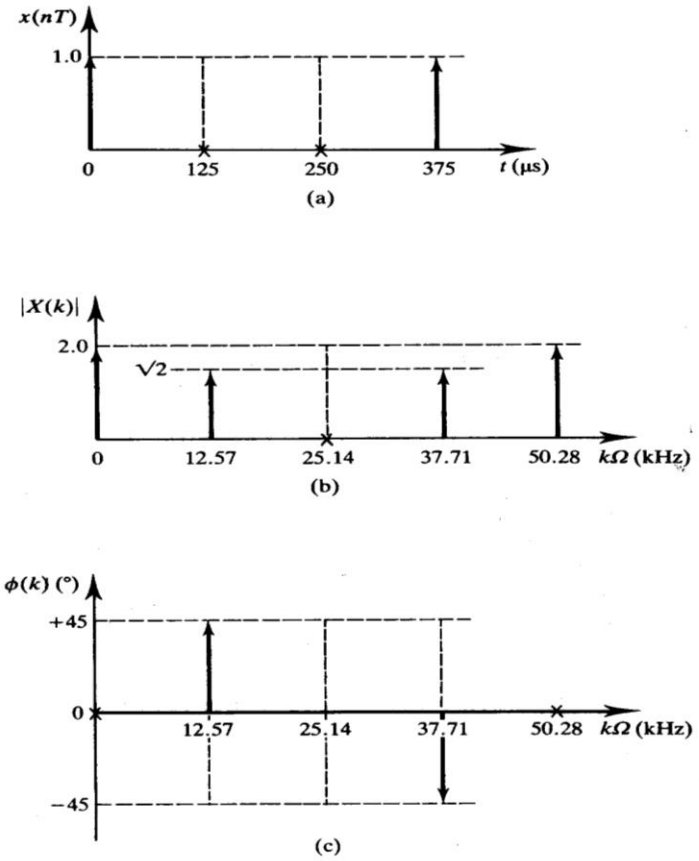


Fig. 2.1: (a) x(nT) versus t, (b)|X(K)| versus k, and (c) Φ(k) versus k,

Sec. 2.1 Cont. - 11

An important property of the DFT may be deduced if the kth component of the DFT, X(k), is compared with the $(k+N)^{th}$ component, X(k+N). Thus

$$X(k) = \sum_{n=0}^{N-1} x(nT)\, e^{-jk\Omega nT}$$

$$= \sum_{n=0}^{N-1} x(nT)\, e^{-jk\pi\, n/N}$$

and

$$X(k+N) = \sum_{n=0}^{N-1} x(nT)\, e^{-jk\pi\, n/N} \cdot e^{-jN2\pi\, n/N}$$

$$= \sum_{n=0}^{N-1} x(nT)\, e^{-jk\pi\, n/N} \cdot e^{-j2\pi\, n}$$

$$= \sum_{n=0}^{N-1} x(nT)\, e^{-jk\pi\, n/N} = X(k)$$

since n is integral so $e^{-j\,2\pi\, n} = 1$

Sec. 2.1 Cont. - 12

The fact that X (k + N) = X(k) shows that the DFT is periodic with period N.

This is the cyclical property of the DFT. The values of the DFT components are repetitive. If k =0, then k + N = N and X(0) = X(N). In the above example X(0) = 2 and therefore X(4) = 2 also. This is illustrated in Figure 2.1(b) where the fourth harmonic amplitude is drawn at 50.28 kHz.

The symmetry of the amplitude distribution about the second harmonic is obvious.

The general conclusion is that the amplitude spectrum of an N-point DFT is symmetrical about harmonic N/2.

If 2 f_{max} samples per second were taken of the signal for t seconds then 2 f_{max} t = N, so

1/t = 2 f_{max} /N is the first harmonic frequency.

The symmetry at harmonic N/2 therefore occurs at the frequency (N/2) (2 f_{max} /N) = f_{max}, the maximum frequency present in the signal.

Sec. 2.1 Cont. - 13

Thus, all the signal components are fully represented in an amplitude spectrum plotted up to f_{max} or harmonic component N/2, and it is unnecessary to plot further points.

In this context, f_{max} is known as the folding frequency, since the spectrum between harmonics N/2 and N may be folded about the axis of symmetry at f_{max} to superimpose exactly the low frequency half of the spectrum.

It is now seen that N real data values transform to N/2 complex DFT values of practical significance. The latter consist of N/2 real values and N/2 imaginary values giving a total of N values derived from the initial N real data values.

Sec. 2.1 Cont. - 14

Finally, the values of the Fourier transform components, F(jω), of the data {1,0,0,1} in Example 2.1 may be obtained by multiplying the DFT components by T = 125 μs. i. e.

$$F(j\omega) = DFT * T$$

Therefore F(0) = 250 μV Hz^{-1},

F(12.57kHz) =(125+j125) μV Hz^{-1} ,

F(25.14kHz) = 0V Hz^{-1} , and

F(37.71kHz)=(125-j125) μV Hz^{-1}

Sec. 2.2

2.2 INVERSE DISCRET FOURIER TRANSFORM (IDFT)

To obtain the discrete transformation from the frequency to the time domain, this achieved by using the IDFT, defined by:

$x(nT) = F_D^{-1}[X(k)]$

$$= \frac{1}{N} \sum_{k=0}^{N-1} X(k)\, e^{jk\Omega nT} \text{ , } n = 0, 1, \ldots\ldots, N-1 \qquad (2.7)$$

where F_D^{-1} denotes the inverse discrete Fourier transformation.

Sec. 2.2 Cont. - 1

The analogy with Eq. F-2 for the inverse Fourier transform is obvious. This time it is quite simple to show that the inverse Fourier transform is obtainable from the IDFT by dividing the IDFT by T.

i. e.

$$\text{IFT} = \text{IDFT} / T$$

The validity of Eq. 2.7 can be demonstrated by substituting for x(nT) in Eq. 2.2.

$$f(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} F(J\omega)\, e^{j\omega t}\, d\omega = \int_{-\infty}^{\infty} F(J\omega)\, e^{j\omega t}\, df \qquad \text{(F-2)}$$

Sec. 2.2 Cont. - 2

Example 2.2

It is useful to illustrate the inverse discrete Fourier transform by using it to derive the time series {1,0,0,1} from its DFT components [2,1+j,0,1-j].

Solution:

With n = 0,

x(nT) = x(0)

$$= \frac{1}{N} \sum_{k=0}^{N-1} X(k)$$

= ¼ [X(0) + X(1) + X(2) + X(3)]

= ¼ [2 + (1+j) + 0 + (1-j)]

= 1, as expected.

Sec. 2.2 Cont. - 3

With n = 1,

$$x(nT) = x(T)$$

$$= \frac{1}{N} \sum_{k=0}^{N-1} X(k)\, e^{jk\Omega T}$$

$$= \frac{1}{N} \sum_{k=0}^{N-1} X(k)\, e^{jk2\pi/N}$$

$$= \tfrac{1}{4} \sum_{k=0}^{N-1} X(k)\, e^{jk\pi/2}$$

$$= \tfrac{1}{4} [2 + (1 + j)\, e^{j\pi/2} + 0 + (1 - j)\, e^{j3\pi/2}]$$

$$= \tfrac{1}{4} [2 + (1 + j)j + (1 - j)(-j)]$$

$$= \tfrac{1}{4} (2 + j - 1 - j - 1)$$

$$= 0 \text{ , as expected.}$$

Sec. 2.2 Cont. - 4

With n = 2,

$$x(nT) = x(2T)$$

$$= \frac{1}{N} \sum_{k=0}^{N-1} X(k)\, e^{jk\Omega T.2} \quad , \qquad \Omega = \frac{2\pi}{NT}$$

$$= \frac{1}{4} \sum_{k=0}^{3} X(k)\, e^{jk\frac{2\pi}{NT}..T2}$$

$$= \frac{1}{4} \sum_{k=0}^{3} X(k)\, e^{jk(\frac{4\pi}{N})} \quad , \qquad N = 4$$

$$= \frac{1}{4} \sum_{k=0}^{3} X(k)\, e^{jk\pi}$$

$$= \frac{1}{4} [2 + (1 + j)\, e^{j\pi} + (1 - j)\, e^{j3\pi}]$$

$$= \frac{1}{4} [2 - (1 + j) - (1 - j)]$$

$$= \frac{1}{4} [2 - 1 - j - 1 + j]$$

= 0, as expected.

Sec. 2.2 Cont. - 5

With n = 3,

$$x(nT) = x(3T) = = \frac{1}{4}\sum_{k=0}^{3} X(k)\, e^{jk\Omega 3T}$$

$$= \frac{1}{4}\sum_{k=0}^{3} X(k)\, e^{jk\frac{2\pi}{4NT}.T.3}$$

$$= \frac{1}{4}\sum_{k=0}^{3} X(k)\, e^{j\frac{3k\pi}{2}}$$

$$= \frac{1}{4}\sum_{k=0}^{3} X(k)\, e^{jk(\frac{3\pi}{2})}$$

$$= \frac{1}{4}[2 + (1 + j)\, e^{j\frac{3\pi}{2}} + (1 - j)\, e^{j\frac{9\pi}{2}}]$$

$$= \frac{1}{4}[2 + (1 + j)(-j) + (1 - j)j]$$

$$= \frac{1}{4}(2 - j + 1 + j + 1]$$

$= 1$, as expected.

the correct final term of the series.

Sec. 2.3

2.3 PROPERTIES OF THE DFT

The DFT has a number of mathematical properties which can be used to simplify problems or which lead to useful applications. Some of them are listed below.

(1) Symmetry:

$$\mathrm{Re}[X(N-k)] = \mathrm{Re}\, X(k) \qquad (2.8)$$

Where: Re…. denotes the real part

states the symmetry of the amplitude spectrum, and

$$\mathrm{Im}\ [X(N-k)] = -\,\mathrm{Im}\,[X(k)] \qquad (2.9)$$

Where: Im…. denotes the imaginary part

states the antisymmetrical property of the phase spectrum.

This is a property to be aware of when considering component values.

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Sec. 2.3 Cont. - 1

(2) Even functions:

If x(n) is an even function xe(n), i.e. xe(n) = xe(-n), then

$$FD[X_e(n)] = X_e(k) = \sum_{n=0}^{N-1} X_e(n) \cos(k\Omega nT) \quad (2.10)$$

(3) Odd functions:

If x(n) is an odd function x0(n), i.e. x0(n) = -x0(-n), then

$$FD[X_O(n)] = X_O(k) = -j\sum_{n=0}^{N-1} X_O(n) \sin(k\Omega nT) \quad (2.11)$$

Sec. 2.3 Cont. - 2

(4) Perceval's Theorem:

The normalized energy in the signal is given by either of the expressions:

$$x2(n) = \sum_{n=0}^{N-1} X^2(n) = \frac{1}{N} \sum_{n=0}^{N-1} |X(K)|^2 \qquad (2.12)$$

the right-hand side is the mean square spectral amplitude, while

the left-hand side is the sum of the squared magnitudes of the time series**.**

(5) Delta function: $FD[(\delta nT)] = 1$ (2.13)

(6) Linear cross-correlation of two data sequences or series may be computed using DFTs.

(7) DFTs may also be used in the computation of circular convolutions, and, by using augmenting zeros, in linear convolutions.

Sec. 2.4

2.4 COMPUTATIONAL COMPLEXITY OF THE DFT

A large number of multiplications and additions are required for the calculation of the DFT. For an 8-point DFT the expansion for X(k) becomes (from Eq. 2.4).

$$X(k)=\sum_{n=0}^{7} x(n)e^{-\frac{2\pi jkn}{8}}, \qquad k=0,\ldots\ldots,7 \qquad (2.14)$$

and letting $\frac{k2\pi}{8} = K$ **this expands to:**

$$X(k) = x(0)e^{-jk0} + x(1)e^{-jk1} + x(2)e^{-jk2} + x(3)e^{-jk3} + x(4)e^{-jk4} + x(5)^{-jk5} + x(6)e^{-jk6} + x(7)e^{-jk7},$$

$$k=0,1, ..7 \qquad (2.15)$$

Sec. 2.4 Cont. - 1

$$X(k) = x(0)e^{-jk0} + x(1)e^{-jk1} + x(2)e^{-jk2} + x(3)e^{-jk3} + x(4)e^{-jk4} + x(5)^{-jk5} + x(6)e^{-jk6} + x(7)e^{-jk7},$$

$k=0,1, ..7$ (2.15)

Each of the product terms is added together. There are therefore eight complex multiplications and seven complex additions to be calculated. For N-point DFT, there will be N and N - 1 of them respectively.

There are also eight harmonic components to be evaluated (k = 0, . . ., 7). This number becomes N for an N -point DFT.

So, for an N-point DFT these complex multiplications and complex additions to be calculated become N^2 and N(N-1) respectively.

Sec. 2.4 Cont. - 2

Numbers complex multiplications and complex additions in DFT is summarized in the following table:

Operation	DFT
No. of Complex Addition	N (N-1)
No. of Complex Multiplications	N^2

If N = 1024, then approximately one million complex multiplications and one million complex additions are required.

Clearly some means of reducing these numbers is desirable.

For example,

if k =1 and n=2, so,

$$e^{\frac{-2\pi jkn}{8}} = e^{\frac{-j\pi}{2}}, \text{ and}$$

if k =2 and n =1,

$$e^{\frac{-2\pi jkn}{8}} = e^{\frac{-j\pi}{2}}, \text{ also}$$